(Revised) Regulation- 2020 (PCC)

Reg. No: | | | | | | | | | | | |

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS257 - HIGH PERFORMANCE COMPUTING

Answer ALL questions

Duration: 2 Hrs 30 min **Maximum: 70 Marks**

PART - A **(5x 1 = 5 Marks)**

1. **CO1** Which network topology provides the shortest average path length for communication in parallel computers?
   a) Hypercube b) Mesh c) Ring d) Star

2. **CO2** Which partitioning strategy is used in the DNS algorithm of matrix multiplication?
   a) 1D partition b) 2D partition c) 3D partition d) Block partition

3. **CO3** What feature enables a kernel to launch other kernels in a parallel computing environment?
   a) Warp shuffle b) Grid-stride loop
   c) Texture memory d) CUDA dynamic parallelism

4. **CO3** Which of the following CUDA functions is used to allocate memory on the GPU device?
   a) cudaMalloc() b) malloc() c) cudaMallocHost() d) mallocHost()

5. **CO2** What does MPI provide to define communication layouts among processes in a non-physical layout?
   a) Message tagging b) Derived data types
   c) Virtual topologies d) MPI-IO

PART - B **(5x 3 = 15 Marks)**

6. **CO1** Define multicore and multithreaded processors with one key difference.
7. **CO1** What is the significance of memory hierarchy in modern processors?
8. **CO2** What is meant by data scoping in OpenMP?
9. **CO2** List any three differences between Point-to-Point communication and Collective communication in MPI.
10. **CO3** What is meant by data locality in GPU programming?

PART - C **(5 x 10 = 50 Marks)**

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) List and explain any four applications of parallel programming in real-world domains. (5)

    CO1 ii) Differentiate Static and Dynamic Mapping technique for load balancing. (5)

12. **CO1** Explain term of all-to-all broadcast on linear array, mesh & Hypercube topologies. (10)

13. **CO2** Explain Parallelism in Best First Search Algorithm. Give an Appropriate example. (10)

14. **CO2** i) Write short note on circular shift on a mesh. (5)

    CO2 ii) Discuss the issues in sorting for parallel computers. (5)

15. **CO3** i) Explain the synchronization behavior of CUDA memory copy APIs. (5)

    CO3 ii) Explain different kinds of CUDA memory. (5)

16. **CO1** The following table shows the number of instructions of a program. (10)

Arithmetic	Store	Load	Branch	Total
500	50	100	50	700

- Assuming that arithmetic instructions take 1cycle, load and store take 5 cycles and branch takes 2 cycles. What is the execution time of a program in 2GHz processor?

- Find the cpi of the program.
- If the number of load instructions can be reduced to half, what is the speed up and cpi?

17. **CO3** i) Give five applications of CUDA. (5)

    CO3 ii) Design a simple CUDA Kernel function to multiply two integers. (5)

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